

IMPORTANCE OF SKIN

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2025 Edition

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Chapter 1 Importance of Skin

Importance of Skin

Skin is the largest organ of the body, accounting for approximately 15% of total body weight and covering a surface area of roughly 1.5–2 m². It provides an essential living biological barrier against mechanical trauma, ultraviolet radiation, microbial invasion, and chemical irritants. Beyond its protective role, the skin actively regulates body temperature through vasodilation, vasoconstriction, sweating, and insulation by subcutaneous fat. It prevents transepidermal water loss, thereby maintaining systemic hydration and internal homeostasis.

Underlying systemic disease can frequently be heralded by changes on the skin surface, making the skin an invaluable diagnostic window into internal pathology. Dermatological manifestations may precede, accompany, or follow systemic illness, and their recognition can lead to earlier diagnosis and treatment of conditions ranging from autoimmune disease to malignancy.

Close inspection of the entire skin surface, including the nails, hair, and mucous membranes, should be the basis of routine skin examination. This comprehensive approach ensures that subtle but clinically significant findings are not overlooked.

Skin diseases can profoundly affect patients' psychiatric health and overall quality of life. Patients with visible skin disease frequently experience stigma, social isolation, anxiety, depression, and diminished self-esteem. A validated measure of how much skin disease affects patients' lives can be made using the **Dermatology Life Quality Index (DLQI)**, a 10-question scored instrument widely used in clinical practice and research.

KEY TEACHING POINTS

Skin is the largest organ and serves as a barrier, thermoregulator, and water-loss prevention system.

The skin can provide the first clinical clue to underlying systemic disease — always examine the skin thoroughly.

Nails, hair, and mucous membranes must never be omitted during skin examination.

Psychosocial impact of skin disease is substantial — use the DLQI as a validated quality-of-life measure.

The DLQI contains 10 questions covering symptoms, daily activities, leisure, work/school, personal relationships, and treatment burden. Scores range from 0 (no impact) to 30 (extreme impact).

Skin Anatomy

The skin is a complex, multilayered organ composed of three distinct layers, each with unique structural and functional properties. Understanding skin anatomy is fundamental to comprehending the pathophysiology of dermatological disease and interpreting clinical findings.

1. Epidermis

The epidermis is the outermost layer of the skin, composed predominantly of **keratinocytes** (squamous epithelial cells). It is **avascular**, relying on diffusion from the underlying dermal capillaries for its metabolic needs. The epidermis is rich in **keratin**, a structural protein that provides strength and resilience.

The epidermis itself is subdivided into five strata (from deepest to most superficial):

- **Stratum basale (basal layer):** A single layer of columnar or cuboidal keratinocytes that undergo mitosis, continuously replenishing the epidermis. Melanocytes, Merkel cells, and Langerhans cells are also found in this layer.
- **Stratum spinosum (prickle cell layer):** Several layers of polyhedral keratinocytes connected by desmosomes, giving a "spiny" appearance on histology.

- **Stratum granulosum (granular layer):** Three to five layers of flattened keratinocytes containing keratohyalin granules and lamellar bodies, critical for barrier function.
- **Stratum lucidum:** A thin, translucent layer found only in thick skin (palms and soles).
- **Stratum corneum (horny layer):** The outermost layer, consisting of 15–30 layers of dead, flattened, anucleate corneocytes embedded in a lipid matrix — the primary physical barrier of the skin.

The **epidermal turnover time** (the duration for a keratinocyte to migrate from the basal layer to the surface) is approximately 28–40 days. This turnover rate is accelerated in conditions such as psoriasis (reduced to 3–5 days), contributing to the characteristic scaling.

2. Dermis

The dermis is the middle and thickest layer of the skin, composed of connective tissue. It contains:

- Blood vessels (arterioles, venules, capillaries)
- Lymphatic vessels
- Nerves and specialized sensory receptors (Meissner's corpuscles, Pacinian corpuscles)
- Hair follicles
- Sweat glands (eccrine and apocrine)
- Sebaceous glands
- Arrector pili muscles

The dermis provides structural support, elasticity, and tensile strength to the skin. It is divided into the **papillary dermis** (superficial, loose connective tissue with dermal papillae interlocking with the epidermis) and the **reticular dermis** (deeper, dense irregular connective tissue).

3. Subcutaneous Fat (Hypodermis Subcutis)

The subcutaneous layer lies beneath the dermis and is composed primarily of adipose tissue and loose connective tissue. It serves as:

- An energy reservoir
- A thermal insulator
- A mechanical cushion protecting underlying structures
- An anchor for the skin to underlying fascia

KEY TEACHING POINTS

The epidermis is avascular — it relies entirely on the dermis for nutrient supply.

The stratum corneum is the primary physical barrier; its dysfunction underlies many dermatoses.

Epidermal turnover time is ~28 days normally but accelerates dramatically in psoriasis (3–5 days).

The dermis houses all skin appendages — understanding this is critical for understanding wound healing and scar formation.

Subcutaneous fat thickness varies by body region, age, sex, and nutritional status.

Comparison of Skin Layers

Feature	Epidermis	Dermis	Subcutis
Composition	Keratinocytes (squamous epithelium)	Connective tissue, vessels, nerves, appendages	Adose tissue, loose connective tissue
Vascularity	Avascular	Highly vascular	Vascular
Thickness	0.05–1.5 mm	0.3–3.0 mm	Variable (1–50+ mm)
Primary function	Barrier protection	Structural support, sensation	Insulation, cushioning, energy storage
Key cell types	Keratinocytes, melanocytes, Langerhans cells, Merkel cells	Fibroblasts, macrophages, mast cells	Adipocytes

Skin Lesions: Primary and Secondary

A skin lesion refers to any area of skin that has different characteristics from the surrounding skin. The ability to accurately describe and classify skin lesions is the single most important clinical skill in dermatology, as the morphology of a lesion often points directly to the diagnosis.

Skin lesions are divided into two main types:

- **Primary skin lesions:** Originate on previously healthy skin and are directly associated with a specific cause. These are the fundamental building blocks of dermatological diagnosis.
- **Secondary skin lesions:** Develop from the evolution of a primary skin lesion, either due to traumatic manipulation (scratching, rubbing), treatment, or natural disease progression.

MCQ TRAPS & PITFALLS

Do NOT confuse primary and secondary lesions — a patient may have both simultaneously.

A lesion that starts as a vesicle but is scratched may become an excoriation or erosion — always look for the primary lesion.

Secondary changes (scaling, crusting, lichenification) can obscure the primary lesion — examine carefully for the earliest lesion.

Primary Skin Lesions

Macules and Patches

Flat lesions with changes in color but no elevation above the surface of the surrounding skin. They represent altered pigmentation or erythema without any palpable component.

- **Macule:** <1 cm in diameter — e.g., freckle, café-au-lait spot, early flat mole
- **Patch:** >1 cm in diameter — e.g., vitiligo, port-wine stain, large café-au-lait macule

Key distinguishing feature: the lesion is flat and cannot be felt with the fingertip. Diascopy (pressing a glass slide against the lesion) can help differentiate erythema (blanches) from purpura (does not blanch).

Papules and Plaques

Raised, solid lesions that can be felt above the surface of the surrounding skin.

- **Papule:** <1 cm in diameter — e.g., mole, insect bite, acne lesion
- **Plaque:** >1 cm in diameter — e.g., psoriasis, lichen planus, eczema plaque

Plaques may form from the coalescence of multiple papules. The surface of a plaque may be smooth, scaly, or verrucous.

Nodule

A palpable, circumscribed lesion with greater volume than a papule. Nodules are >1 cm in diameter and involve the dermis and/or the subcutaneous tissue (subcutis). They feel firm or rubbery on palpation.

- Examples: dermatofibroma, epidermoid cyst, lipoma, erythema nodosum, metastatic skin nodules

Nodules differ from papules in their depth of involvement — they extend into the dermis or subcutis rather than being confined to the superficial skin.

Vesicles and Bullae

Fluid-filled lesions (blisters) containing clear fluid.

- **Vesicle:** <1 cm in diameter — e.g., chickenpox (varicella), herpes simplex, contact dermatitis
- **Bulla:** >1 cm in diameter — e.g., bullous pemphigoid, bullous impetigo, burns, pemphigus vulgaris

The location of the blister within the skin (intraepidermal vs. subepidermal) is diagnostically critical and can be distinguished by the **Nikolsky sign** — lateral pressure on the blister causes extension in intraepidermal blistering disorders (positive Nikolsky sign, as in pemphigus vulgaris).

Pustules

Small, circumscribed lesions filled with pus (purulent fluid containing neutrophils). Pustules may arise from infection or sterile inflammatory processes.

- **Infectious causes:** Impetigo, folliculitis, acne vulgaris
- **Sterile causes:** Pustular psoriasis, pustular drug eruption, rosacea

Wheal

A transient elevation of the skin caused by dermal edema. Wheals are characteristically:

- Pale centrally with an erythematous flare at the margin
- Intensely pruritic (itchy)
- Transient — individual lesions typically resolve within 24 hours
- A component of **urticaria** (hives)

Wheals represent a type I hypersensitivity reaction with mast cell degranulation and histamine release. The transient nature of individual lesions is a key diagnostic feature.

Maculopapular Rash

A collection of small skin lesions that are a mixture of flat macules and raised papules. This is one of the most common rash morphologies encountered in clinical practice.

Common causes include:

- Drug eruptions (morbilliform drug eruption)
- Scarlet fever
- Secondary syphilis
- Rubella and other viral exanthems
- Graft-versus-host disease

Secondary Skin Lesions

Lichenification

Hard thickening of the skin with accentuated skin markings (exaggeration of normal skin lines). It commonly results from chronic inflammation and repeated rubbing or scratching of the skin.

- Classic example: lichenification in chronic eczema / atopic dermatitis
- The skin appears leathery, thickened, with prominent furrows

Lichenification is a sign of chronicity — it indicates that the patient has been scratching or rubbing the area over a prolonged period.

Scale

Hyperkeratosis — accumulation of stratum corneum due to increased proliferation and/or delayed desquamation of keratinocytes. Scales appear as peeling or flaking of the skin surface.

- **Examples:** Psoriasis (silvery-white scales), lichen planus, eczema, tinea infections, ichthyosis
- Scale color and character can be diagnostically helpful: silvery-white in psoriasis, greasy yellow in seborrheic dermatitis

Crust

Dried serum, blood, or pus on the surface of the skin. Crusts (also called scabs) form when exudate from a weeping or erosive lesion dries on the skin surface.

- **Example:** Impetigo (honey-colored crusts)

Fissure

Linear slits through the full thickness of the skin. Fissures are usually painful and occur in areas of skin flexion or where the skin is dry and thickened.

- **Examples:** Angular cheilitis (fissures at the corners of the mouth), hand dermatitis, heel fissures

Erosion

A superficial loss of epidermis that generally heals without scarring. Erosions are shallow, moist, and pink or red in appearance.

- **Examples:** Impetigo, friction blisters after rupture, trauma, pemphigus

The key distinction from an ulcer: erosions involve only the epidermis and heal without scarring.

Ulcer

Loss of the full thickness of the epidermis and upper dermis. Healing results in a scar. Ulcers are deeper than erosions and may be painful.

- **Examples:** Stasis ulcer (venous ulcer), pyoderma gangrenosum, arterial ulcer, pressure ulcer (decubitus ulcer), diabetic foot ulcer

Excoriations

Partial or complete loss of epidermis as a result of scratching. Excoriations are linear or angular erosions caused by mechanical trauma from the patient's nails.

- **Examples:** Neurotic excoriations, acne excoriée, prurigo nodularis

Atrophy

Loss of tissue affecting the epidermis, dermis, or subcutaneous fat. Epidermal atrophy is characterized by thinning of the skin with loss of normal skin markings, fine wrinkles, loss of pigment, and a translucent appearance. Dermal atrophy may show sclerosis of underlying connective tissue, telangiectasia, or diminished blood supply.

- **Examples:** Lichen sclerosus, skin aging (senile atrophy), poikiloderma, striae (stretch marks), corticosteroid side effect (iatrogenic atrophy)

Desquamation

Peeling of superficial scales, often occurring during the healing phase of a skin condition or as a primary feature of certain diseases.

- **Examples:** Post-streptococcal desquamation (scarlet fever), keratolytic peeling in Kawasaki disease, physiological desquamation in newborns

Comprehensive Lesion Classification Table

Primary vs. Secondary Skin Lesions — Summary

Lesion Type	Morphology	Size	Examples
PRIMARY LESIONS			
Macule	Flat, color change only	<1 cm	Freckle, café-au-lait spot
Patch	Flat, color change only	>1 cm	Vitiligo, port-wine stain
Papule	Raised, solid	<1 cm	Mole, insect bite

Lesion Type	Morphology	Size	Examples
Plaque	Raised, solid, flat-topped	>1 cm	Psoriasis, lichen planus
Nodule	Palpable, deep (dermis/subcutis)	>1 cm	Lipoma, dermatofibroma
Vesicle	Fluid-filled blister	<1 cm	Herpes simplex, varicella
Bulla	Fluid-filled blister	>1 cm	Bullous pemphigoid, pemphigus
Pustule	Pus-filled	Variable	Acne, impetigo, pustular psoriasis
Wheal	Transient, edematous, pale center	Variable	Urticaria
Maculopapular	Mixed flat and raised lesions	Variable	Drug eruption, viral exanthem
SECONDARY LESIONS			
Lichenification	Thickened skin, accentuated markings	Variable	Chronic eczema
Scale	Hyperkeratosis, flaking	Variable	Psoriasis, tinea
Crust	Dried serum/blood/pus	Variable	Impetigo
Fissure	Linear full-thickness slit	Variable	Angular cheilitis
Erosion	Superficial epidermal loss	Variable	Pemphigus, trauma
Ulcer	Full-thickness loss (epidermis + dermis)	Variable	Stasis ulcer, pyoderma gangrenosum
Excoriation	Scratch-induced epidermal loss	Variable	Prurigo, neurotic excoriations
Atrophy	Tissue loss, thinning	Variable	Lichen sclerosus, steroid use
Desquamation	Peeling of superficial scales	Variable	Scarlet fever (convalescent phase)

Clinical Approach to Skin Examination

A systematic approach to skin examination is essential for accurate diagnosis. The following framework should be applied to every patient:

- 1. Distribution:** Where are the lesions? (localized, generalized, symmetrical, asymmetrical, sun-exposed areas, flexural, extensor)
- 2. Configuration:** How are the lesions arranged? (annular, linear, grouped, dermatomal, reticular, targetoid)
- 3. Morphology:** What is the primary lesion type? (use the classification above)
- 4. Color:** What color are the lesions? (erythematous, hyperpigmented, hypopigmented, violaceous, flesh-colored)

5. Border: Are the borders well-demarcated or poorly defined?

6. Secondary changes: Is there scaling, crusting, lichenification, ulceration, or scarring?

OSCE PEARLS

Always examine the entire skin surface, including the scalp, nails, hair, and mucous membranes — pathology may be hidden.

Use a magnifying lens or dermatoscope for detailed examination of pigmented lesions.

Perform diascopy (press a glass slide against erythematous lesions) to distinguish erythema (blanches) from purpura (does not blanch).

Check the Nikolsky sign in any patient with blisters — lateral pressure causing extension suggests intraepidermal blistering (pemphigus).

Document lesions with clear descriptions using standard dermatological terminology — this is essential for monitoring and communication.

Practice Questions

Q: A 45-year-old man presents with well-demarcated, erythematous, silvery-scaled plaques on his elbows and knees. The plaques have been present for several months and are slowly enlarging. On examination, you also notice pitting of the fingernails. What is the most likely diagnosis?

- A. Atopic dermatitis
- B. Psoriasis
- C. Lichen planus
- D. Seborrheic dermatitis
- E. Tinea corporis

Answer: B

Explanation: The key features pointing to psoriasis are: (1) well-demarcated erythematous plaques with silvery-white scales; (2) extensor surface distribution (elbows and knees); (3) chronic, slowly progressive course; (4) nail pitting, which is a characteristic associated feature of psoriasis. Atopic dermatitis typically affects flexural surfaces with less well-demarcated lesions. Lichen planus presents with violaceous, polygonal, pruritic papules. Seborrheic dermatitis has greasy yellow scales in seborrheic areas. Tinea corporis shows annular lesions with central clearing.

Q: A 6-year-old child presents with grouped vesicles on an erythematous base on the trunk and face. Some vesicles have ruptured, leaving erosions with honey-colored crusts. The child has a low-grade fever. What is the most likely diagnosis?

- A. Varicella (chickenpox)
- B. Herpes simplex
- C. Bullous impetigo
- D. Contact dermatitis

E. Scabies

Answer: C

Explanation: The key features pointing to bullous impetigo are: (1) vesicles that have ruptured leaving erosions; (2) characteristic honey-colored crusts; (3) perioral and truncal distribution; (4) low-grade fever. Varicella shows lesions at different stages (papules, vesicles, crusts) simultaneously ("starry sky" pattern). Herpes simplex typically affects the lips or genital area. Contact dermatitis is pruritic with vesicles but lacks honey-colored crusts. Scabies shows burrows and affects web spaces, wrists, and axillae.

Q: A 32-year-old woman presents with a 2-day history of raised, itchy, pale-centered lesions with erythematous margins on her arms and trunk. Each lesion resolves within a few hours but new lesions continue to appear. What is the most likely diagnosis?

- A. Erythema multiforme
- B. Urticaria
- C. Maculopapular drug eruption
- D. Viral exanthem
- E. Lichen planus

Answer: B

Explanation: The key features pointing to urticaria (hives) are: (1) raised, itchy wheals with pale centers and erythematous margins; (2) transient nature — individual lesions resolve within 24 hours; (3) new lesions appearing in different locations. Erythema multiforme shows target lesions that are fixed and do not resolve within hours. Maculopapular drug eruptions and viral exanthems are not transient — individual lesions persist for days. Lichen planus presents with violaceous polygonal papules that are persistent.

Q: A 55-year-old man with a history of chronic venous insufficiency presents with a shallow, painful ulcer on the medial aspect of his left ankle. The ulcer has a granulation tissue base and irregular borders. The surrounding skin shows hemosiderin deposition and varicose veins. What is the most likely diagnosis?

- A. Arterial ulcer
- B. Diabetic foot ulcer
- C. Venous stasis ulcer
- D. Pyoderma gangrenosum
- E. Pressure ulcer

Answer: C

Explanation: The key features pointing to a venous stasis ulcer are: (1) location on the medial ankle (gaiter area) — the classic site; (2) shallow ulcer with granulation tissue base; (3) history of chronic venous insufficiency; (4) surrounding hemosiderin deposition and varicose veins. Arterial ulcers are typically on the toes and lateral malleolus with a punched-out appearance. Diabetic foot ulcers occur on pressure points of the foot. Pyoderma gangrenosum has violaceous undermined borders. Pressure ulcers occur over bony prominences.

Q: A 28-year-old woman presents with hypopigmented, flat, well-demarcated patches on her hands and face that have been slowly enlarging over the past year. The patches are completely asymptomatic. On Wood's lamp examination, the patches show bright white fluorescence. What is the most likely diagnosis?

- A. Pityriasis versicolor
- B. Vitiligo
- C. Post-inflammatory hypopigmentation
- D. Tuberous sclerosis (ash-leaf spots)
- E. Leprosy

Answer: B

Explanation: The key features pointing to vitiligo are: (1) hypopigmented (not depigmented initially) well-demarcated patches; (2) progressive enlargement over time; (3) asymptomatic; (4) bright white fluorescence on Wood's lamp (indicating complete melanocyte loss). Pityriasis versicolor shows fine scaling and is caused by Malassezia. Post-inflammatory hypopigmentation follows a preceding inflammatory condition. Tuberous sclerosis ash-leaf spots are present from childhood and are associated with other systemic features. Leprosy causes hypopigmented anesthetic patches.